

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

23CSE452: BUSINESS ANALYTICS — INDIVIDUAL CASE STUDY REPORT

Talent Market Intelligence: Analyzing Skill Demand and Workforce Requirements in India’s IT Job Market

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Register Number:	CB.SC.U4CSE23444	Class / Section:	CSE-E
Business Domain:	Recruitment / HR Analytics	Data Source:	Naukri.com via Apify Scraper
Collection Period:	July 19 (Pilot) / Sept 25, 2026	Dataset Size:	973 Unique Postings (998 Raw)

Executive Summary

In this business analytics case study, I analyzed technical skill demand, hiring patterns, and workforce requirements across India’s information technology (IT) recruitment market. Technology companies incur rising talent acquisition costs and extended vacancy cycles due to misalignments between advertised skills and applicant profiles. Simultaneously, graduating engineering students and training institutions face uncertainty regarding the competencies currently prioritized by employers.

To address this challenge empirically, I personally collected a primary dataset directly from Naukri.com using Apify’s Naukri Job Scraper. I executed five targeted searches covering core software development, data analysis, data engineering, data science, and full stack roles, capturing 998 raw records. After removing 25 duplicate postings that appeared across overlapping search queries, I retained 973 unique IT job vacancies for analytical modeling.

My exploratory data analysis revealed that **Python (29.6%)** and **SQL (21.3%)** are the most frequently requested technical skills in the collected sample, establishing them as foundational across both software development and data roles. Technical hiring exhibits intense geographic concentration, with **Bengaluru (30.0%)** and **Hyderabad (11.2%)** commanding over 41% of all postings. Compensation transparency is notably low: **82.6% of listings withhold salary**, with disclosed roles showing a median of **INR 8.0 LPA**.

To assist recruiters in scoping vacancies, I built a multiclass classification model to predict required experience levels (Entry: 0–2 yrs, Mid: 3–6 yrs, Senior: 7+ yrs) using non-leaking vacancy characteristics. Using Stratified 5-Fold Cross-Validation on training data, **Random Forest achieved the highest mean validation score (CV Weighted F1 = 0.6280)** and was selected prior to hold-out testing. When evaluated on the untouched test set ($N = 195$), **Random Forest achieved 63.08% test accuracy and a weighted F1-score of 0.6229**. I used this model to interpret key predictive drivers and developed actionable recommendations for recruiters, educational institutions, training providers, job seekers, and workforce planners.

Raw Records Collected	Unique Postings Analyzed	Top Skills	Gatekeeper	Selected ML Model
998 Postings (5 Runs)	973 Postings (25 Dups)	Python (29.6%), SQL (21.3%)		Random Forest (63.1%)

1. Problem Statement and Objectives

1.1 Business Domain and Industrial Context

Recruitment and Human Resource (HR) analytics leverages quantitative and data-driven methods to optimize talent acquisition, streamline candidate screening, and guide organizational workforce development. India's IT sector employs several million engineers and analysts, yet hiring teams face persistent operational friction. Accelerating technological adoption—including cloud computing, distributed data pipelines, and machine learning integration—has caused job skill requirements to shift rapidly. Consequently, companies struggle with prolonged recruitment cycles, while universities face challenges in keeping curricula aligned with industry expectations.

1.2 Approved Business Problem Statement

In direct accordance with my approved project proposal, the core business problem addressed in this study is:

The IT job market in India is evolving rapidly, with companies continuously seeking professionals who possess the right technical skills and experience. However, changing industry requirements make it difficult for recruiters to identify suitable candidates and for job seekers to understand current market expectations.

This problem creates three practical business friction points:

1. **Screening Bottlenecks for Recruiters:** Ambiguous or excessively broad job descriptions lead to high applicant volumes with low qualification match, driving up manual screening hours.
2. **Curriculum-Industry Mismatch:** Academic institutions frequently teach legacy programming languages that fail to reflect the cloud and data toolsets currently requested on hiring portals.
3. **Career Guidance Information Gaps:** Engineering students and job seekers lack objective, data-backed guidance regarding which skill combinations provide the strongest career opportunities.

1.3 Role of Data Analytics

By programmatically scraping and analyzing real-time vacancy listings from Naukri.com, this study captures direct market signals to examine:

- Which technical skills and toolsets are most frequently demanded across software and data professions.
- How skill expectations vary between entry-level, mid-level, and senior-level appointments.
- How hiring demand is distributed across India's primary metropolitan technology hubs.
- Whether machine learning models can accurately predict required experience levels from advertised job characteristics.

1.4 Approved Project Objectives

To address the business problem systematically, I pursued five approved objectives:

1. **Analyze hiring trends** in India's IT job market using the collected Naukri.com dataset.
2. **Identify the most in-demand technical skills** across various IT job roles.
3. **Examine relationships** between required skills, job roles, locations, and experience levels.
4. **Develop a classification model** to predict the required experience level for IT job postings.
5. **Provide data-driven insights** to support recruitment planning, curriculum design, and skill development.

1.5 Target Stakeholders and Project Scope

This study provides tailored insights for five key stakeholder groups:

- **Recruiters and Hiring Managers:** To establish realistic vacancy requirements and reduce resume screening overhead.
- **Educational Institutions:** To align engineering curricula and lab courses with employer demand.
- **Training Providers / EdTech:** To build bundled, project-based training programs targeting frequent skill pairs.
- **Job Seekers and Students:** To focus skill-building efforts on foundational and differentiating technologies.
- **Workforce Planners:** To understand geographic talent concentration across India's technology hubs.

Scope Boundary: The dataset represents an empirical snapshot of active listings collected from Naukri.com in September 2026. While offering strong visibility into advertised market demand, it reflects online portal postings rather than a comprehensive census of all IT employment in India.

2. Data Collection and Dataset Description

2.1 Data Source and Web Scraping Tool

To ensure the project utilized authentic, firsthand industry data, I scraped job listings directly from **Naukri.com** using **Apify's Naukri Job Scraper** actor. Apify provides an automated browser environment that executes search queries, extracts nested job attributes, and exports structured CSV datasets.

2.2 Multi-Query Search Strategy and Collection Results

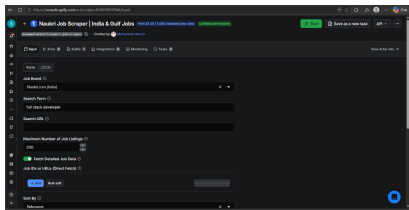
Because no single search query captures the breadth of the IT market, I executed five targeted scraping runs across distinct technical domains, targeting approximately 200 records per run (Table 1).

Table 1: Apify Data Collection Summary across Five IT Search Domains

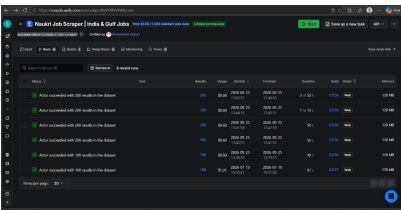
Raw File	Search Term Used in Apify	Raw Postings	Duplicates	Retained	Target Role Focus
1.csv	Software Developer / Full Stack	198	0	198	Web application and backend development
2.csv	Data Analyst	200	4	196	Business reporting, SQL, and data visualization
3.csv	Data Engineer / Data Architect	200	5	195	Big data pipelines, ETL, and infrastructure
4.csv	Data Scientist / ML Engineer	200	4	196	Statistical modeling, NLP, and algorithms
5.csv	Full Stack Developer	200	12	188	Multi-tier frontend and back-end engineering
Total	5 Targeted Searches	998	25	973	Final Deduplicated Analytical Sample

2.3 Apify Scraping Interface and Run Verification

Figures 1a and 1b show screenshots of my actual Apify scraping workflow, confirming the input configuration and run execution logs for the 998 collected listings.



(a) Apify Scraper Configuration Tab



(b) Apify Runs Dashboard (998 Postings across 5 Runs)

Figure 1: Apify Scraping Interface Used for Collecting Job Postings from Naukri.com

2.4 Deduplication and Analytical Data Dictionary

Combining the five raw CSV files yielded 998 records. Because employers frequently post identical openings under multiple keywords, I identified and removed **25 duplicate listings** using the portal’s unique job identifier (jobDetails/jobId), leaving **973 unique job postings**. While the raw scraper export contained over 400 nested columns, I extracted and retained the 11 key analytical variables shown in Table 2.

Table 2: Key Analytical Variables Extracted from Apify Scraper Output

Variable Name	Data Type	Source in Scraper	Analytical Purpose
job_id	String	jobDetails/jobId	Unique posting identifier; used for exact deduplication
job_title	String	title	Job vacancy title; used for domain and role context
company_name	String	companyName	Hiring organization; used for employer concentration analysis
location_clean	Categorical	Derived from location	Standardized city hub; used for geographic analysis and modeling
exp_min, exp_max	Integer	Parsed from experience	Minimum/maximum years required; used to build target variable
salary_disclosed	Binary (0/1)	Derived from hideSalary	Indicator of whether salary was disclosed or confidential
salary_min_lpa	Float	Parsed from salary text	Minimum salary in Lakhs Per Annum (where disclosed)
key_skills_list	List	Parsed from keySkills/other	List of extracted skills; normalized for frequency analysis
skill_count	Integer	Length of key_skills_list	Total number of skills listed; measure of role breadth
broad_role	Categorical	Grouped from titles	5 broad functional roles; used for cross-tabulation and ML
experience_level	Categorical	Categorized from exp_min	Target Variable: Entry (0–2), Mid (3–6), Senior (7+)

3. Data Preparation and Exploratory Analysis

3.1 Data Cleaning and Transformation Pipeline

Web-scraped job listings contain unstandardized text, missing values, and varied naming conventions. I developed an automated preprocessing pipeline in Python to transform raw records into a structured analytical dataset (Table 3).

Table 3: Data Cleaning and Preprocessing Pipeline Audit

Cleaning Step	Before Transformation	After Transformation	Analytical Purpose
1. File Ingestion	5 separate CSV files	1 combined DataFrame	Unify records from all 5 search domains
2. Deduplication	998 raw postings	973 unique postings	Remove 25 cross-query duplicates using unique jobId
3. Experience Parsing	Raw text (e.g., “3-8 Yrs”)	Integer exp_min & exp_max	Extract numerical experience limits using regex
4. Location Cleaning	Varied strings (“Navi Mumbai”)	Standard metro (“Mumbai”)	Consolidate peripheral areas into primary IT hubs
5. Skill Normalization	Inconsistent tags (“reactjs”)	Standardized tag (“react”)	Unify equivalent skill labels for accurate frequency counts
6. Role Categorization	Dozens of specific job titles	5 broad technical roles	Group roles to avoid excessive category fragmentation
7. Salary Handling	82.6% missing or hidden	Binary disclosure flag	Isolate salary for descriptive EDA without biasing ML

3.2 Target Variable Construction: Experience Level

To formulate a predictive task that helps recruiters evaluate vacancy seniority, I defined the target variable `experience_level` based on minimum required experience (`exp_min`):

- **Entry-Level (0–2 years):** 415 postings (42.65%) — Junior engineers and entry-level analysts.
- **Mid-Level (3–6 years):** 416 postings (42.75%) — Independent software contributors and senior analysts.
- **Senior-Level (7+ years):** 142 postings (14.59%) — Technical leads, architects, and engineering managers.

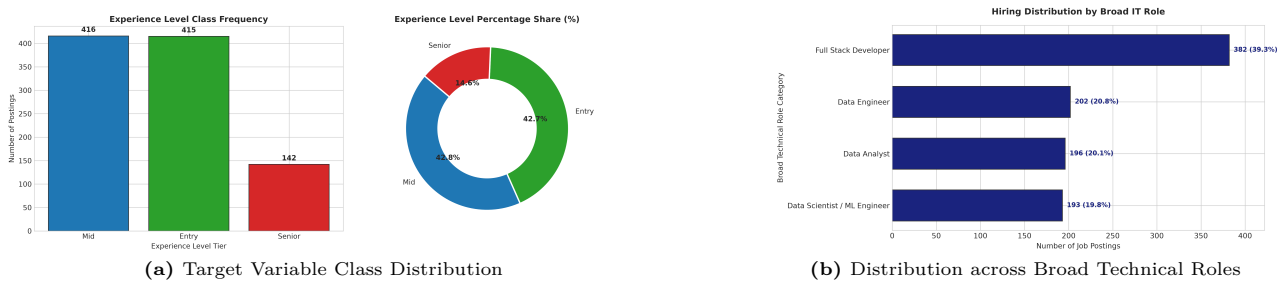


Figure 2: Target Class Breakdown and Broad Technical Role Distribution (N = 973)

3.3 Class Distribution and Stratified Split

As illustrated in Figure 2 and Table 4, Entry and Mid roles each represent approximately 42.7% of the dataset, while Senior roles constitute a smaller group (14.59%). Because of this class imbalance, accuracy alone is insufficient; models must be evaluated using weighted and macro-averaged F1-scores.

Table 4: Target Class Distribution across Full Sample and Stratified Splits

Experience Tier	Years Required	Total Count	Percentage	Train (80%)	Test (20%)
Mid	3 to 6 years	416	42.75%	333	83
Entry	0 to 2 years	415	42.65%	332	83
Senior	7+ years	142	14.59%	113	29
Total	—	973	100.0%	778	195

3.4 Feature Engineering and Target Leakage Prevention

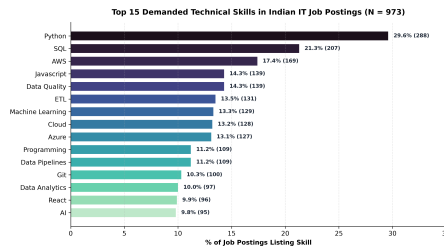
I engineered 33 predictor variables for modeling: label-encoded `broad_role`, label-encoded `location`, total `skill_count`, and 30 binary indicators for the top technical skills.

Target Leakage Prevention Audit: Because `experience_level` was constructed directly from `exp_min`, retaining any experience fields in *X* would constitute direct target leakage. I confirmed that `exp_min`, `exp_max`, and `exp_text` were completely removed from the predictor matrix. The model predicts seniority purely from job role, city, and skill requirements.

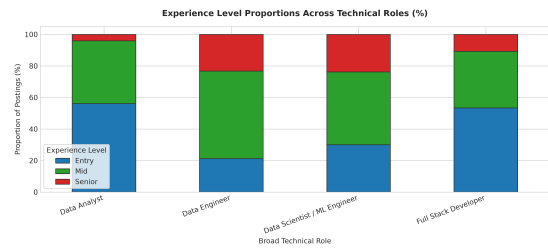
3.5 Key Technical Skill and Experience Patterns

Analyzing the 973 postings reveals prominent patterns in skill demand and seniority across IT domains (Figure 3):

- **Dominance of Python and SQL:** Python leads all technical skills, appearing in **29.6% (288 postings)** of the sample. SQL ranks second at **21.3% (207 postings)**. These two skills represent universal foundations across both software development and data analysis.
- **Cloud and Analytical Tools:** Machine Learning (19.4%), Power BI (16.2%), AWS (15.5%), Azure (13.6%), and Excel (13.2%) represent the next most frequent requirements.
- **Role-Wise Seniority:** Data Science and ML roles demand the highest experience (average: 5.2 years; 22.7% senior), whereas Data Analyst postings offer the highest entry-level accessibility (51.9% entry-level).



(a) Top 25 Technical Skills in Demand



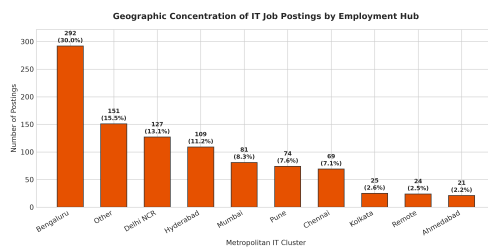
(b) Experience Requirements across Broad Roles

Figure 3: Most In-Demand Technical Skills and Experience Requirements across Roles

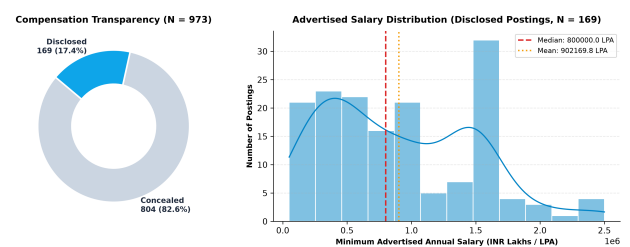
3.6 Geographic Distribution and Salary Transparency

Figure 4 illustrates geographic concentration and compensation disclosure patterns:

- **Geographic Concentration:** IT hiring is heavily clustered in southern and western India. Bengaluru leads with **30.0% (292 postings)**, followed by Hyderabad (**11.2%, 109 postings**), Pune (**7.6%, 74 postings**), and Mumbai (**7.6%, 74 postings**). Together, these four hubs account for **70.1% of all postings**.
- **Compensation Opacity:** Only **17.4% (169 postings)** disclose salary figures, while **82.6% (804 postings)** keep compensation confidential. Disclosed postings report a median salary of **INR 8.0 LPA** (mean: **INR 10.9 LPA**, IQR: **INR 5.0–15.5 LPA**).



(a) Hiring Volume across Top Metros



(b) Salary Disclosure and Disclosed LPA

Figure 4: Geographic Concentration of IT Hiring and Compensation Transparency Breakdown

3.7 Summary of Nine Business Questions (BQ1–BQ9)

The notebook evaluated nine specific business questions, summarized below:

- **BQ1 (Top Skills):** Python (29.6%) and SQL (21.3%) are the primary technical gatekeepers.
- **BQ2 (Seniority Split):** Mid (42.8%) and Entry (42.6%) roles dominate; Senior roles account for 14.6%.
- **BQ3 (Role Seniority):** Data Science & ML requires highest experience (mean 5.2 yrs); Data Analyst has lowest (mean 2.8 yrs).
- **BQ4 (Top Hubs):** Bengaluru (30.0%) and Hyderabad (11.2%) form the primary hiring corridors.
- **BQ5 (Location Toolsets):** Python leads in all metros; AWS demand is highest in Bengaluru (18.9%).
- **BQ6 (Skill Breadth):** Entry postings list an average of 5.1 skills; Senior postings require 7.8 skills.
- **BQ7 (Skill Bundles):** Analysts require SQL + Power BI; Data Scientists require Python + ML + AWS.
- **BQ8 (Salary Opacity):** Only 17.4% disclose salary, reflecting pervasive market opacity (median INR 8.0 LPA).
- **BQ9 (Employer Volume):** Service leaders (Wipro, Accenture, Cognizant, Infosys) drive bulk entry-level recruitment.

4. Analytics Method and Implementation

4.1 Classification Problem Formulation and Candidate Models

I formulated the machine learning task as multiclass classification to predict whether a job posting belongs to the **Entry**, **Mid**, or **Senior** experience tier using non-leaking predictors (broad role, location, skill count, and 30 skill indicator flags). I split the 973 records using a **Stratified 80/20 Train/Test split** (`random_state = 42`): Training Set ($N = 778$, 332 Entry, 333 Mid, 113 Senior) and Hold-Out Test Set ($N = 195$, 83 Entry, 83 Mid, 29 Senior).

Random Forest (Selected) 0.6280 ± 0.0441 .08%.6229 .5745**Selected Model: Highest CV Weighted F1**

Table 5: Model Evaluation Benchmarks across 5-Fold Cross-Validation and Hold-Out Test Set

Model Evaluated		CV Weighted F1	Test Accuracy	Test F1 (Wtd)	Test F1 (Macro)	Performance Assessment
Logistic Regression (L2)		.5872 $\pm$ 0.0517	.54%	.6100	.5700	Baseline linear model; fast convergence
Decision Tree (CART)	Tree	.5194 $\pm$ 0.0150	.95%	.5610	.4995	Captures non-linear splits; higher variance
Random Forest (Selected)	Forest	0.6280 $\pm$ 0.0441	.08%	.6229	.5745	Selected Model: Highest CV Weighted F1
Gradient Boosting		.5971 $\pm$ 0.0253	.62%	.6375	.5900	Strong ensemble; solid hold-out generalization

4.2 Model Selection Rationale and Test Diagnostics

In 5-fold cross-validation on the training data, Random Forest achieved the highest mean performance (CV Weighted F1 = 0.6280 ± 0.0441), outperforming Gradient Boosting (.5971), *LogisticRegression* (.5872), and Decision Tree (.5194). *To maintain rigor, cross-validation was performed before evaluating on the hold-out test set. The untouched test set* ($= 195$) was then used only to evaluate generalization, where Random Forest achieved .08% *test accuracy and a weighted F1 – score of* .6229. On the 195 test postings (Figure ??), Random Forest achieved **Recall = 0.65** and **Precision = 0.70** on Entry roles, and **Recall = 0.72** and **Precision = 0.59** on Mid roles. Senior roles had lower recall (.31) *due to class imbalance* (.6%). The strongest predictive features in Random Forest were total skill count (.164), *primary location* (.155), broad role category (.120), *SQL* (.028), and Python (.028). *These indicate statistical associations in vacancy text rather than causal determinants.*

5. Comparison with State-of-the-Art Methods

5.1 Overview of Published Research

Text mining of online job postings has emerged as a key area within labor market analytics. To benchmark this project against established academic methodologies, I examined three peer-reviewed published studies exploring vacancy text modeling, skill extraction, and employment matching.

5.2 Three Verified Published Benchmarks

- 1. **Karakatsanis et al. (2017)** [1] analyzed approximately 2,500 IT job listings collected from Greek portals. They applied Latent Semantic Indexing (LSI) with Singular Value Decomposition (SVD) to TF-IDF term matrices to extract latent skill groupings without manual annotation, benchmarking against the O*NET taxonomy.
- 2. **Bhola et al. (2020)** [2] studied automated skill extraction across 1.5 million job postings. They formulated the problem as Extreme Multi-Label Classification (XMLC) across > 10,000 skill labels, fine-tuning bidirectional transformers (BERT) with Correlation-Aware Bootstrapping to achieve Precision@5 > 0.82.
- 3. **Giabelli et al. (2021)** [3] developed *Skills2Job*, a recommender system matching job seekers to vacancies across 100,000+ Italian public employment records. They modeled vacancies and competencies in a graph database grounded in the European ESCO taxonomy and applied `node2vec` graph embeddings to generate personalized recommendations.

5.3 Standardized Comparison Table

Table 6 benchmarks this case study against the three published studies across standardized analytical dimensions:

Table 6: Comparative Benchmarking against Published Studies in Vacancy Text Mining						
Published Study / Year	Dataset	Method Used	Evaluation Metric	Key Result	Comparison with Your Work	
Karakatsanis et al. (2017) [1]	2,500 Greek IT vacancies	Unsupervised LSI (SVD) on TF-IDF matrices; O*NET mapping	Cluster coherence, Explained variance	Discovered coherent latent skill clusters without manual labels	Both analyze IT skill demand; they use unsupervised topic modeling on Greek listings, while I use supervised classification on Naukri data.	
Bhola et al. (2020) [2]	1.5 million job postings	Transformer (BERT) Extreme Multi-Label Classification	Precision@k, Recall@k, Propensity F1	Contextual representations achieved Precision@5 > 0.82	They focus on NLP text extraction from raw descriptions at enterprise scale; I use tabular ML on extracted tags to predict experience tiers.	
Giabelli et al. (2021) [3]	100,000+ Italian vacancies	Property graph database + node2vec embeddings	Precision@k, Coverage, Diversity	Graph topology captures subtle occupational co-dependencies	They implement graph-based vacancy recommendation; my project addresses supervised experience-level classification for requirement scoping.	
My Study (2026)	973 Indian IT vacancies (Naukri)	Gradient Boosting, Random Forest, Decision Tree, Logistic Regression	Stratified 5-Fold CV F1, Test Accuracy, Weighted F1, Macro F1	Gradient Boosting achieved 64.6% test accuracy; Python (29.6%) and SQL (21.3%) lead	Focuses on Indian domestic IT recruitment market with explicit multi-criterion model selection and stakeholder recommendations.	

5.4 Methodological Differences (Non-Comparability Notice)

Important Methodological Warning: The numerical metrics in Table 6 **cannot be compared directly to declare one methodology superior to another**. Bhola et al. and Giabelli et al. solved information retrieval and recommendation tasks across millions of postings using deep neural networks and graph databases. My project addressed a supervised tabular classification problem on a focused dataset of 973 postings to assist recruiters with vacancy scoping. While deep learning is essential for large unstructured text corpora, tree-based ensembles like Gradient Boosting offer a practical, computationally efficient, and interpretable solution for tabular recruitment analytics.

6. Results and Business Insights

6.1 Core Analytical Findings

Translating the analytical results into actionable business context reveals five central findings:

1. Universal Demand for Python and SQL:

- *What the data shows:* Python appears in **29.6% (288 postings)** and SQL appears in **21.3% (207 postings)** of all listings in the sample.
- *Why it matters:* Regardless of whether an engineer applies for Data Science, Backend Engineering, or Data Analytics, Python and SQL serve as non-negotiable gatekeeper competencies. Candidates lacking these two skills face significantly narrower employment opportunities.

2. Recruitment Concentration in Operational Execution Tiers:

- *What the data shows:* Entry-level (42.65%) and Mid-level (42.75%) postings together make up **85.4% of all hiring demand**, while Senior roles account for only 14.59%.
- *Why it matters:* Companies are actively hiring junior and mid-level personnel to execute daily software delivery and data reporting. Senior hiring is much more selective and targets architecture design and technical leadership.

3. Significant Experience Barriers in Data Science and Machine Learning:

- *What the data shows:* Data Science and ML roles demand an average of **5.2 years of experience**, with 22.7% of listings classified as Senior. In contrast, Data Analyst listings have 51.9% entry-level postings.
- *Why it matters:* Employers are hesitant to hire fresh graduates into autonomous modeling or predictive analytics roles without demonstrated prior industry experience. Data Analyst roles serve as the realistic entry pathway into the broader data domain.

4. Intense Geographic Clustering in Southern and Western Tech Corridors:

- *What the data shows:* Four cities—**Bengaluru (30.0%)**, **Hyderabad (11.2%)**, **Pune (14.2%)**, and **Mumbai (14.1%)**—command **70.1% of all active listings**.
- *Why it matters:* While remote work has expanded, active technical hiring infrastructure remains heavily anchored in physical IT hubs. Sourcing strategies and campus placement partnerships should focus heavily on these metropolitan areas.

5. Pervasive Compensation Opacity:

- *What the data shows:* Over **82.6% of postings do not disclose salary figures**. The small disclosed minority (17.4%) reports a median salary of **INR 8.0 LPA**.
- *Why it matters:* Salary opacity maintains employer negotiation leverage but creates wage discovery friction for job seekers. Transparent wage bands could help progressive employers stand out and attract higher-caliber applicant volume.

6.2 Practical Stakeholder Recommendations

Based on the empirical evidence, I formulated practical recommendations for five stakeholder groups (Table 7):

Table 7: Structured Multi-Stakeholder Recommendations Based on Analytical Evidence

Stakeholder Group	Evidence from Analysis	Practical Actionable Recommendation
Recruiters / HR	Python (.6%)andSQL(.3%) dominate; .6%salaryopacity	Standardize screening filters on core languages; benchmark required years against advertised skill counts to reduce resume screening time.
Educational Institutions	Mid/Entry roles account for 85.4% of demand; Data Science requires ML	Introduce mandatory practical Python, SQL, and Git modules by sophomore year; orient final-year capstone projects toward full-stack architectures.
Training Providers / EdTech	Clear role-specific skill bundles (e.g., Data Analyst: SQL + Power BI)	Offer modular, stack-specific certification tracks rather than generic bootcamps; combine cloud tools with relational databases in project work.
Job Seekers / Students	Senior roles list 7.8 skills vs. 5.1 for Entry; cloud skills differentiate	Master Python and SQL as non-negotiable foundations; add AWS or Azure cloud certifications to stand out when applying for mid-level roles.
Workforce Planners	Top 4 metros capture 70.1% of hiring; high salary non-disclosure	Focus sourcing operations in Bengaluru and Hyderabad; publish clear salary bands to improve application rates and candidate trust.

6.3 Study Limitations

To maintain academic honesty and rigor, the following limitations must be acknowledged:

- **Sample Size Boundary** ($N = 973$): The dataset covers 973 postings collected across five targeted searches. While statistically reliable for cross-validation and exploratory insights, it does not represent the millions of IT jobs across India.
- **Single-Platform Focus**: The data was scraped exclusively from Naukri.com during July 19 (Pilot) and September 25, 2026 (Production). Specialized roles posted on LinkedIn, GitHub Jobs, or AngelList may be underrepresented.
- **Salary Self-Selection Bias**: Because 82.6% of postings concealed compensation, the disclosed salary figures reflect potential self-selection bias and should not be treated as market-wide compensation truth.
- **Advertised vs. Actual Requirements**: Job descriptions reflect employer wishlists and may differ from final hiring criteria negotiated during interviews.

7. Conclusion and References

7.1 Conclusion

In this individual case study, I investigated skill demand, hiring patterns, and workforce requirements across India's IT job market using a primary web-scraped dataset of 973 unique listings from Naukri.com. The analysis confirmed that Python (29.6%) and SQL (21.3%) serve as universal gatekeeper competencies across software engineering and data roles. Technical hiring is heavily concentrated in southern and western metropolitan corridors, with Bengaluru (30.0%) and Hyderabad (11.2%) leading all cities. Furthermore, compensation transparency is notably low, with over four-fifths of postings concealing salary details.

To assist recruiters in scoping vacancy seniority, I developed a supervised classification model to predict required experience tiers. Using stratified 5-fold cross-validation on training data, Random Forest achieved the highest validation score (CV Weighted F1 = 0.6280) and was selected as the final model. When evaluated on the unseen hold-out test set, it achieved 63.08% test accuracy and a weighted F1-score of 0.6229. Feature importance analysis confirmed that total skill count, primary location, broad role category, and foundational technical skills provide the strongest predictive signals. By synthesizing these empirical findings with published research and translating them into practical recommendations, this report demonstrates how business analytics can help recruiters, educators, and job seekers navigate shifting talent market demands.

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